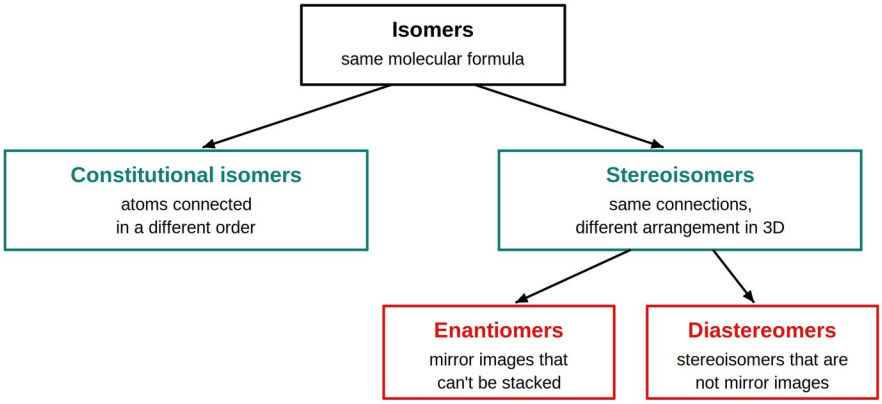
Stereochemistry

Name _____ Date _____

Use these ideas to find chirality centers, assign R and S, and sort pairs of molecules into enantiomers, diastereomers, meso compounds, or the same molecule. Check your R/S answers with the R/S Assigner at chemustad.org.

Isomers: a quick map



Constitutional isomers: Same molecular formula, but the atoms are connected in a different order. Butan-1-ol and butan-2-ol are constitutional isomers.

Stereoisomers: Same atoms, same connections, but a different arrangement in 3D. You cannot turn one into the other without breaking a bond (turning around single bonds doesn't count).

Enantiomers: Stereoisomers that are mirror images of each other but cannot be stacked on top of each other.

Diastereomers: Stereoisomers that are not mirror images. Cis and trans alkenes and rings belong here too.

Your hands are the easiest example. Your left and right hands are mirror images, but a left glove does not fit your right hand no matter how you turn it. A molecule that behaves like that is chiral. A molecule that matches its mirror image, like a plain coffee mug, is achiral.

Same atoms, different smell. The two enantiomers of carvone have the same formula and the same connections, but one smells like spearmint and the other like caraway, the seed in rye bread. The smell receptors in your nose are chiral, so they can tell the two apart, the same way a glove can tell your hands apart.



(R)-carvone
smells like spearmint

(S)-carvone
smells like caraway

Finding chirality centers

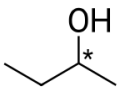
A chirality center (asymmetric center, stereocenter) is usually a carbon with four different groups attached. How I look for them:

1. Skip every CH_2 and CH_3, and every carbon in a double or triple bond. None of them can have four different groups.

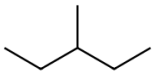
2. For each carbon left, list the four groups. Count H as a group even when it isn't drawn.

3. If two groups start the same way (both CH_2, say), walk out along each one until you find a difference. If you never find one, the groups are the same and the carbon is not a chirality center.

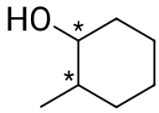
4. In a ring, walk around the ring in both directions from the carbon and compare the two paths the same way.



one chirality center



none: two ethyl groups



two chirality centers

Example: *In 3-methylpentane, C3 holds H, CH_3 and two ethyl groups. The two ethyls are identical, so C3 is not a chirality center. In 2-methylcyclohexan-1-ol, both C1 and C2 are, because the two paths around the ring from each carbon are different.*

Assigning R and S

First rank the four groups (Cahn–Ingold–Prelog rules):

1. Higher atomic number of the atom attached to the stereocenter gets higher priority. Br > Cl > O > N > C > H.

2. If two attached atoms tie, list the three atoms each one is bonded to, highest first, and compare them one at a time. The first difference decides it. Ethyl (C, H, H) beats methyl (H, H, H).

3. A double bond counts as two single bonds to that atom, so C=O counts as C bonded to (O, O).

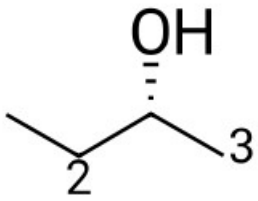
Then read the direction:

1. Put the lowest priority group (usually H) in the back, on a dash.

2. Trace 1 → 2 → 3. Clockwise is R, counterclockwise is S.

3. If the lowest priority group is on a wedge (pointing at you), trace it anyway and flip your answer.

4. Swapping any two groups switches R and S. If the lowest group is in the plane, swap it with the group on the dash, assign, then flip back.

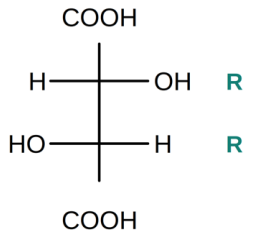


Example: *Butan-2-ol. OH is 1 (highest atomic number). Ethyl beats methyl at the second atom, so ethyl is 2 and methyl is 3. H is 4 and isn't drawn: the OH is on a dash, so the H points toward you. Tracing 1 → 2 → 3 goes counterclockwise, and because H is toward you we flip it: this is (R)-butan-2-ol.*

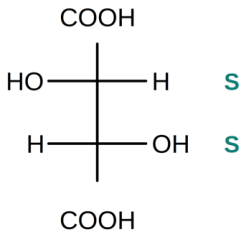
Enantiomers, diastereomers and meso compounds

A molecule with n chirality centers can have up to 2^n stereoisomers. To draw the enantiomer, flip every center: every wedge becomes a dash and every dash a wedge. If you flip some centers but not all, you get a diastereomer.

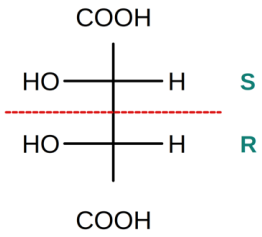
A meso compound has chirality centers but also an internal mirror plane (or a center of symmetry) in at least one conformation, so it is achiral and matches its own mirror image. Tartaric acid has two centers, but only three stereoisomers, because (R,S) and (S,R) are the same meso compound.



(R,R)
chiral



(S,S)
chiral



meso (R,S)
achiral: mirror plane

Fischer projections

A Fischer projection is a shorthand for chains with several stereocenters. Every cross is a carbon. Horizontal lines come toward you (wedges) and vertical lines go away from you (dashes).



Reading R/S: If the lowest priority group is on a vertical line, trace 1 → 2 → 3 as drawn. If it is on a horizontal line (it usually is), trace it and flip the answer.

Moving a Fischer projection: Turning it 180° in the page gives the same molecule. Turning it 90° inverts every center, so you get the enantiomer. Swapping any two groups on one carbon inverts that carbon only.

Optical activity

Enantiomers rotate plane-polarized light by the same amount in opposite directions. A 50:50 mixture (racemic) does not rotate light at all. Enantiomeric excess tells you how far a mixture is from racemic:

ee = % major enantiomer − % minor enantiomer
ee = (observed rotation ÷ specific rotation of the pure enantiomer) × 100

Example: *A mixture of 90% (S) and 10% (R) has 90 − 10 = 80% ee.*

Practice problems

Work through these in order. Check your R/S answers with the R/S Assigner at chemustad.org, and check your 3D picture with a model kit if you have one.

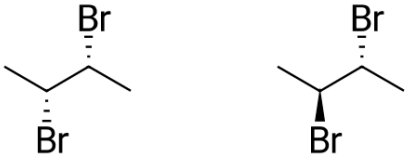
1. Same or different? For each pair, decide whether the two are identical, constitutional isomers, enantiomers, or diastereomers.



(a) _____



(b) _____



(c) _____



(d) _____



(e) _____



(f) _____

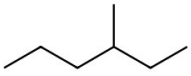


(g) _____

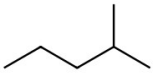


(h) _____

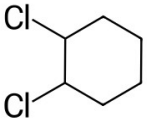
2. Find the chirality centers. Mark every chirality center with an asterisk (*). Some molecules have none.



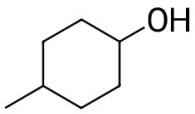
(a)



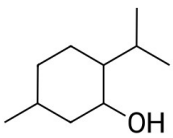
(b)



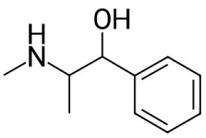
(c)



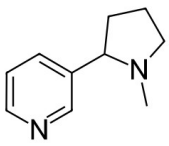
(d)



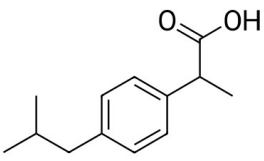
(e)
menthol



(f)
pseudoephedrine

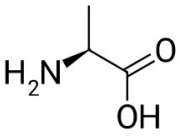


(g)
nicotine

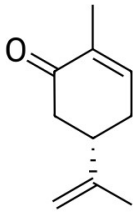


(h)
ibuprofen

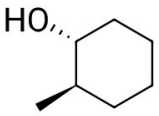
3. Draw the enantiomer. Draw the enantiomer of each molecule below.



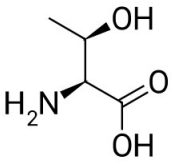
(a)
alanine



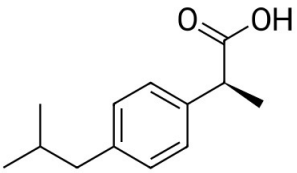
(b)
carvone



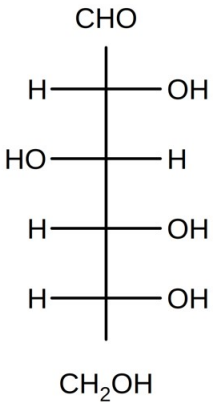
(c)



(d)
threonine

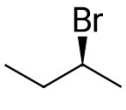


(e)
ibuprofen

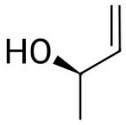


(f)
glucose

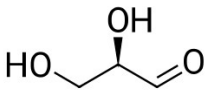
4. Assign R or S. Assign R or S to every chirality center. In (e) the H is drawn on a wedge.



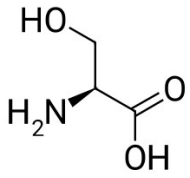
(a)



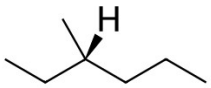
(b)



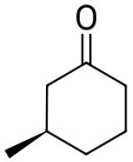
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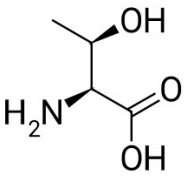
(d)



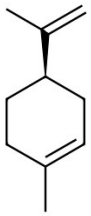
(e)



(f)

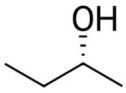


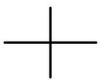
(g)



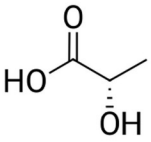
(h)

5. Fischer projections. (a)–(c) Redraw each molecule as a Fischer projection on the cross next to it, with the first carbon of the chain at the top. Then label each center R or S. (d)–(g) Assign R or S to each center in these Fischer projections.



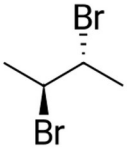


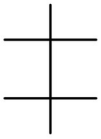
(a)



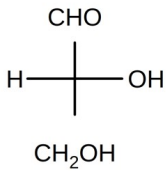


(b)

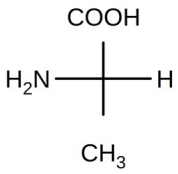




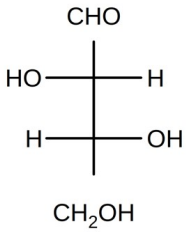
(c)



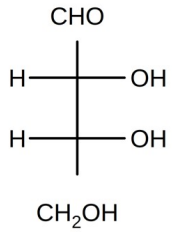
(d)



(e)

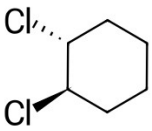


(f)

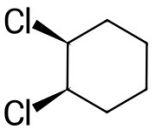


(g)

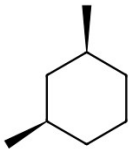
6. Chiral, achiral, or meso? Label each molecule as chiral, achiral (no chirality centers), or meso.



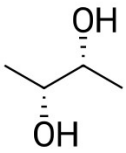
(a)



(b)

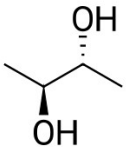


(c)



(d)

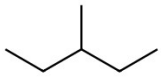
(a) _____ (b) _____ (c) _____ (d) _____



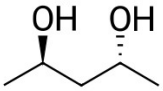
(e)



(f)



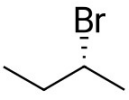
(g)



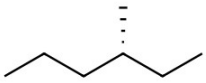
(h)

(e) _____ (f) _____ (g) _____ (h) _____

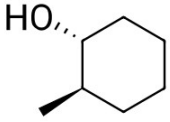
7. Name them. Give the IUPAC name of each compound, including R or S for every chirality center.



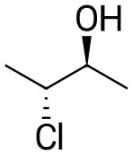
(a)



(b)



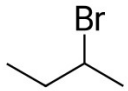
(c)



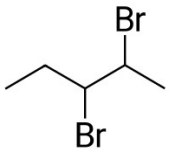
(d)

(a) _____
(b) _____
(c) _____
(d) _____

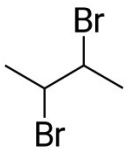
8. How many stereoisomers? How many stereoisomers does each compound have? Watch for meso forms.



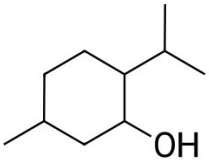
(a)



(b)



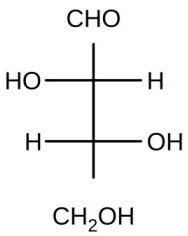
(c)



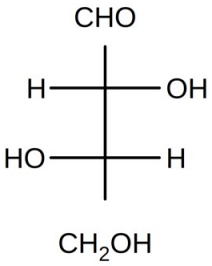
(d)

(a) _____ (b) _____ (c) _____ (d) _____

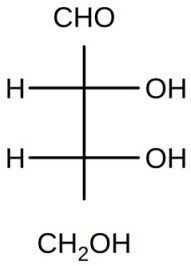
9. Compare to compound A. Give the relationship between compound A and each of (a)–(e): identical, enantiomers, diastereomers, or constitutional isomers.



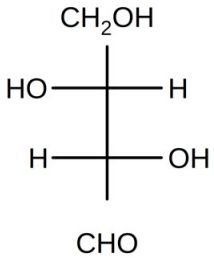
Compound A



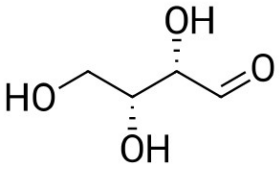
(a)



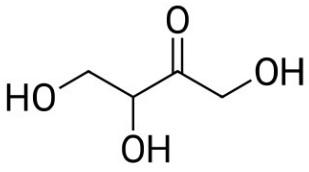
(b)



(c)



(d)



(e)

(a) _____ (b) _____ (c) _____ (d) _____ (e) _____

10. Real samples.

(a) A bottle of carvone contains 80% (R)-carvone and 20% (S)-carvone. What is its ee? What does it mostly smell like?

(b) Ibuprofen is sold as a 50:50 mixture of its two enantiomers. What is its ee, and what optical rotation would you measure?

(c) A sample of a chiral compound has an observed rotation of +30.0°. The pure (S) enantiomer has a rotation of +40.0°. What is the ee, and what percentage of each enantiomer is in the sample?

11. Challenge: see it in 3D. Open the Newman Navigator at chemustad.org and load meso-2,3-dibromobutane from Presets. Look down C2–C3 and turn the back carbon until the two Br atoms are anti. Draw the Newman projection. Where is the symmetry in this conformation, and how does it explain why the meso compound is achiral? Then load (2R,3R)-2,3-dibromobutane and compare.

